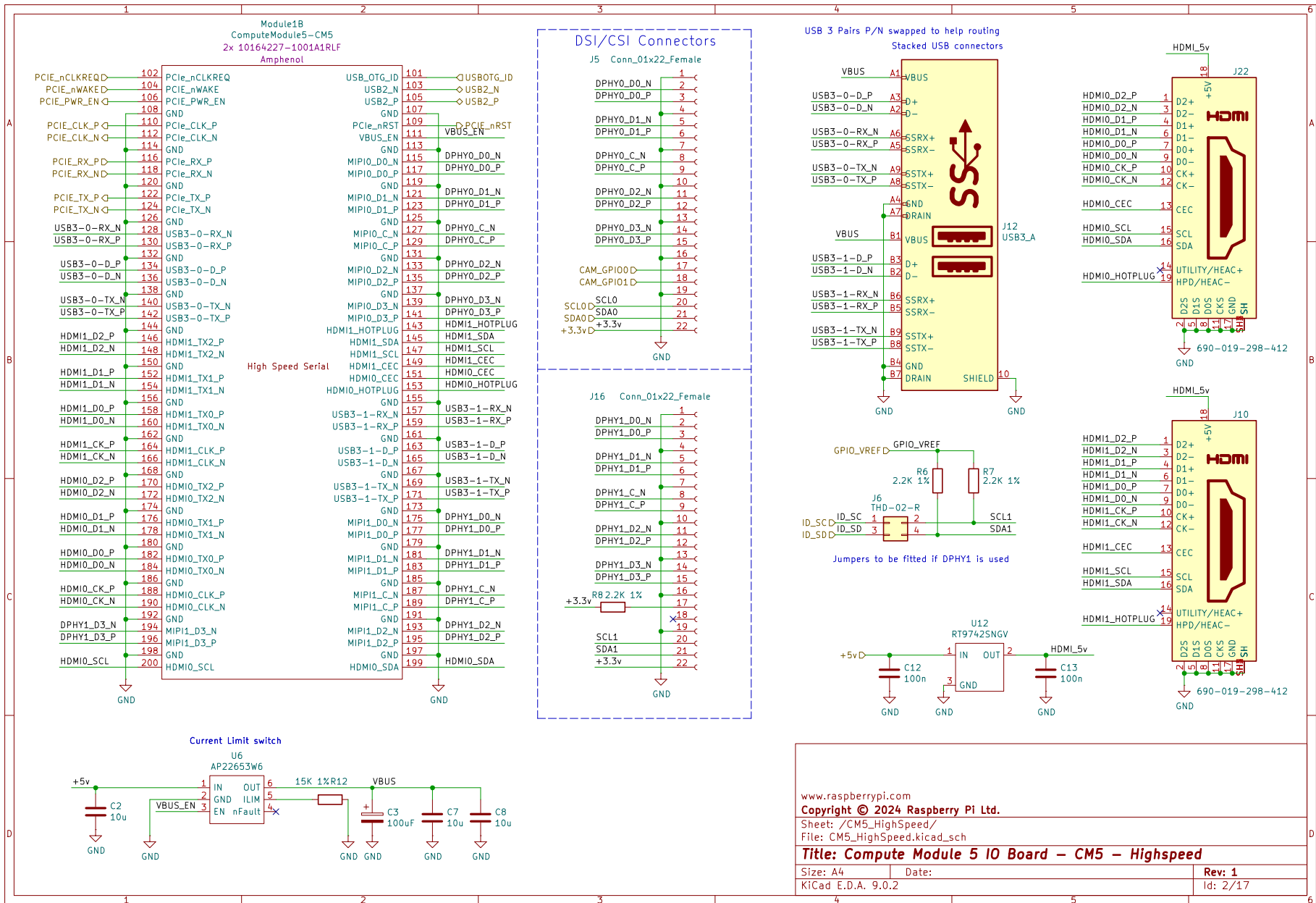
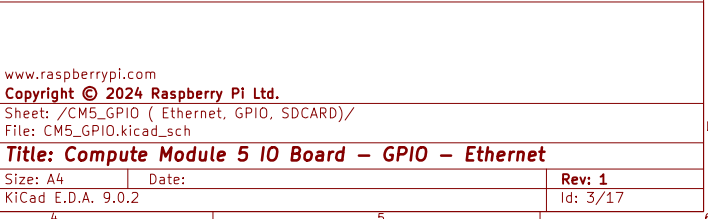
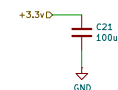
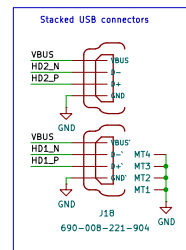
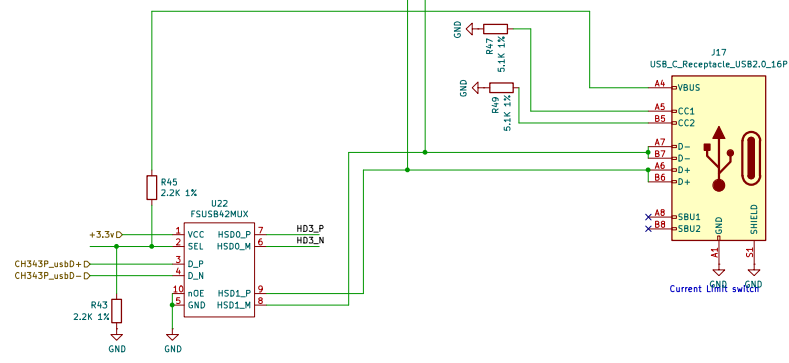
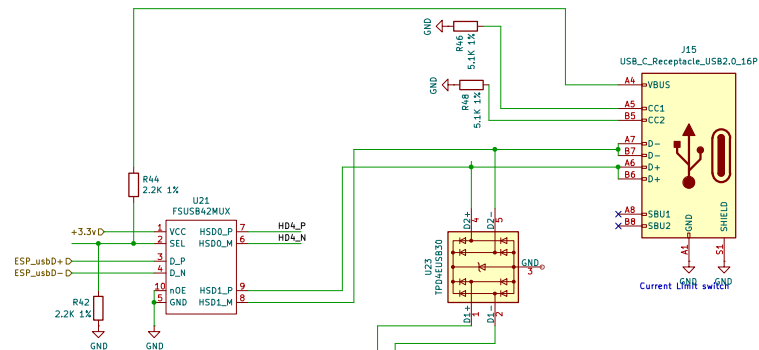
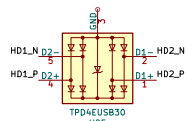
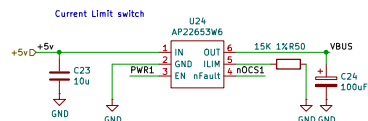
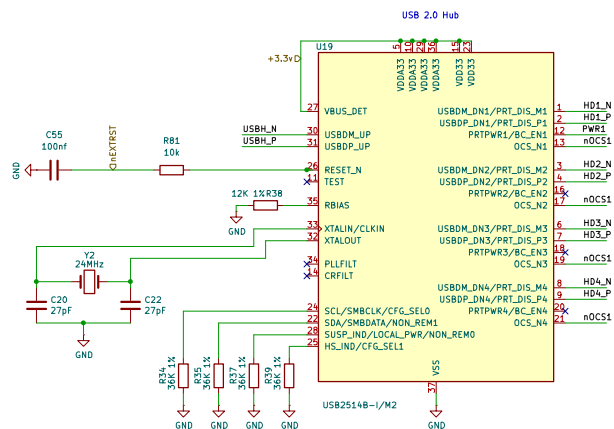
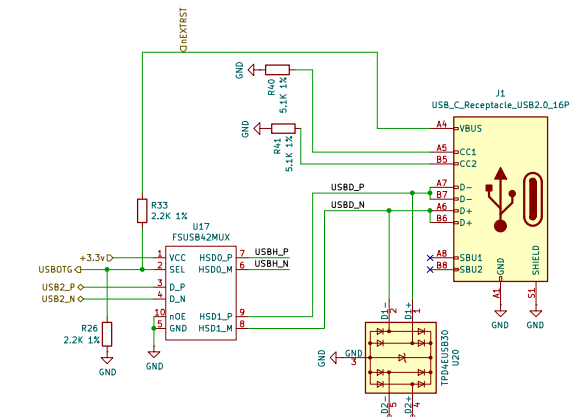
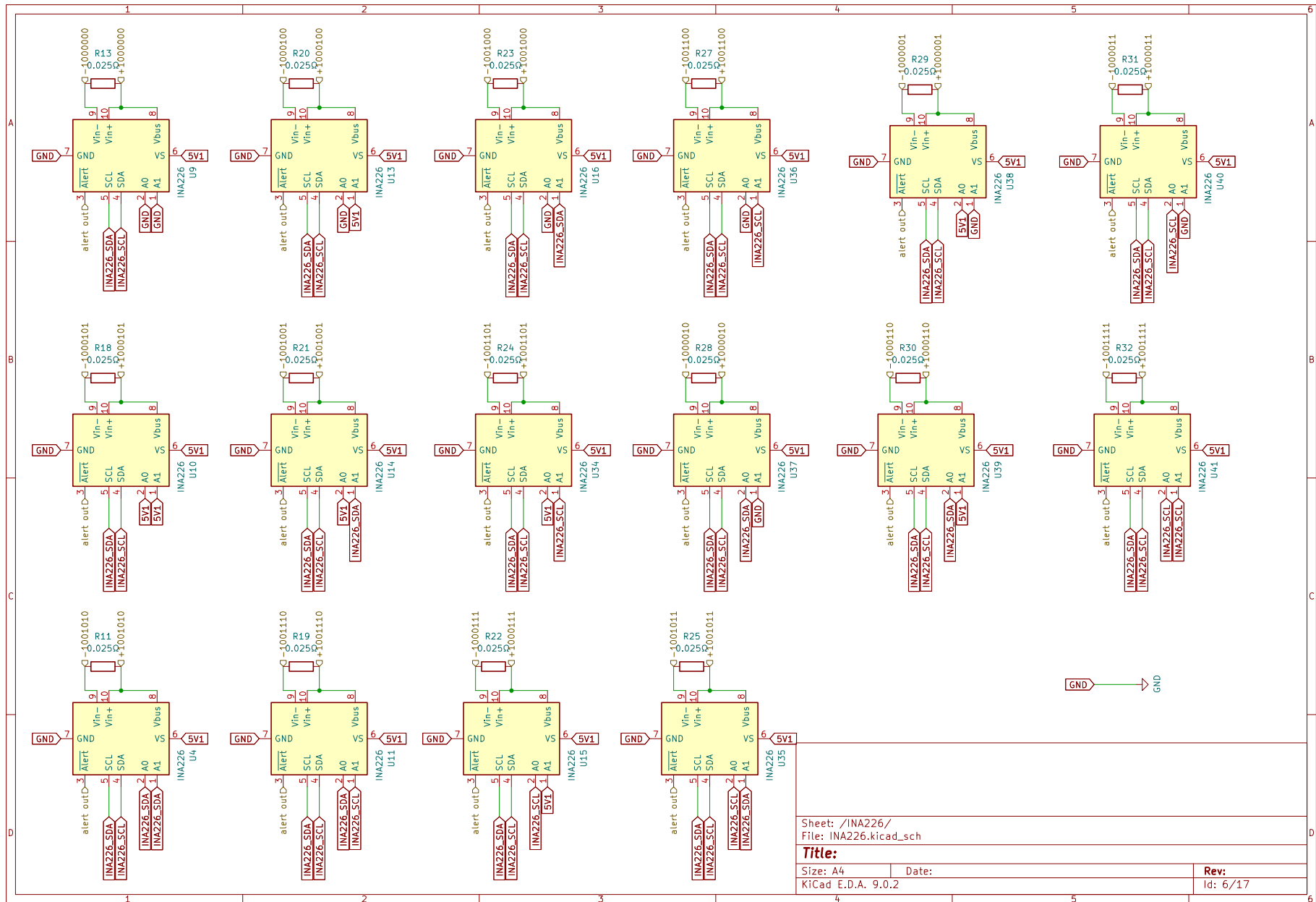
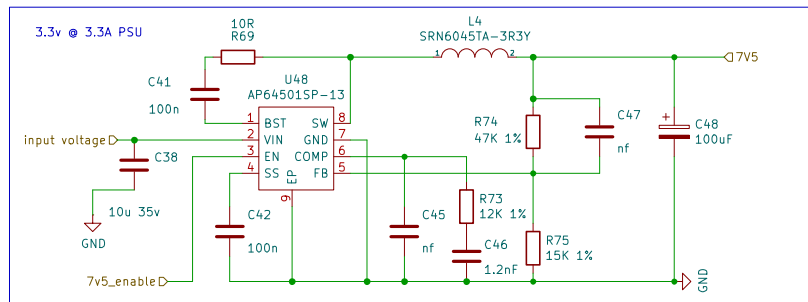
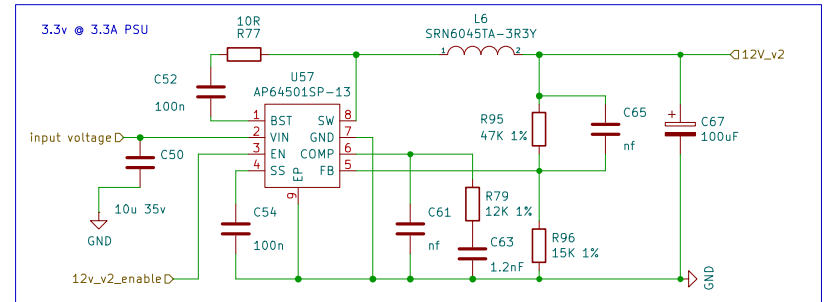
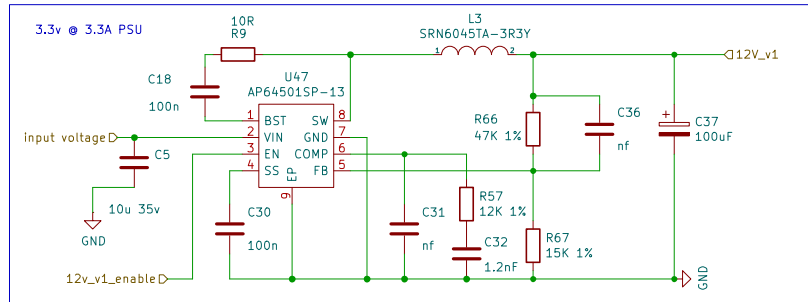
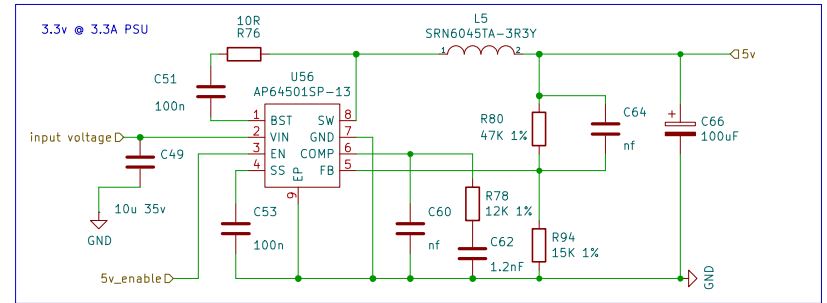
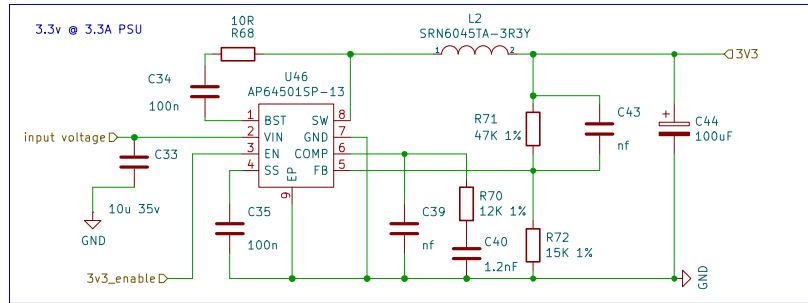












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Sheet: /PSUs/

File: PSUs.kicad_sch

Title: Compute Module 4 IO Board – PSUs

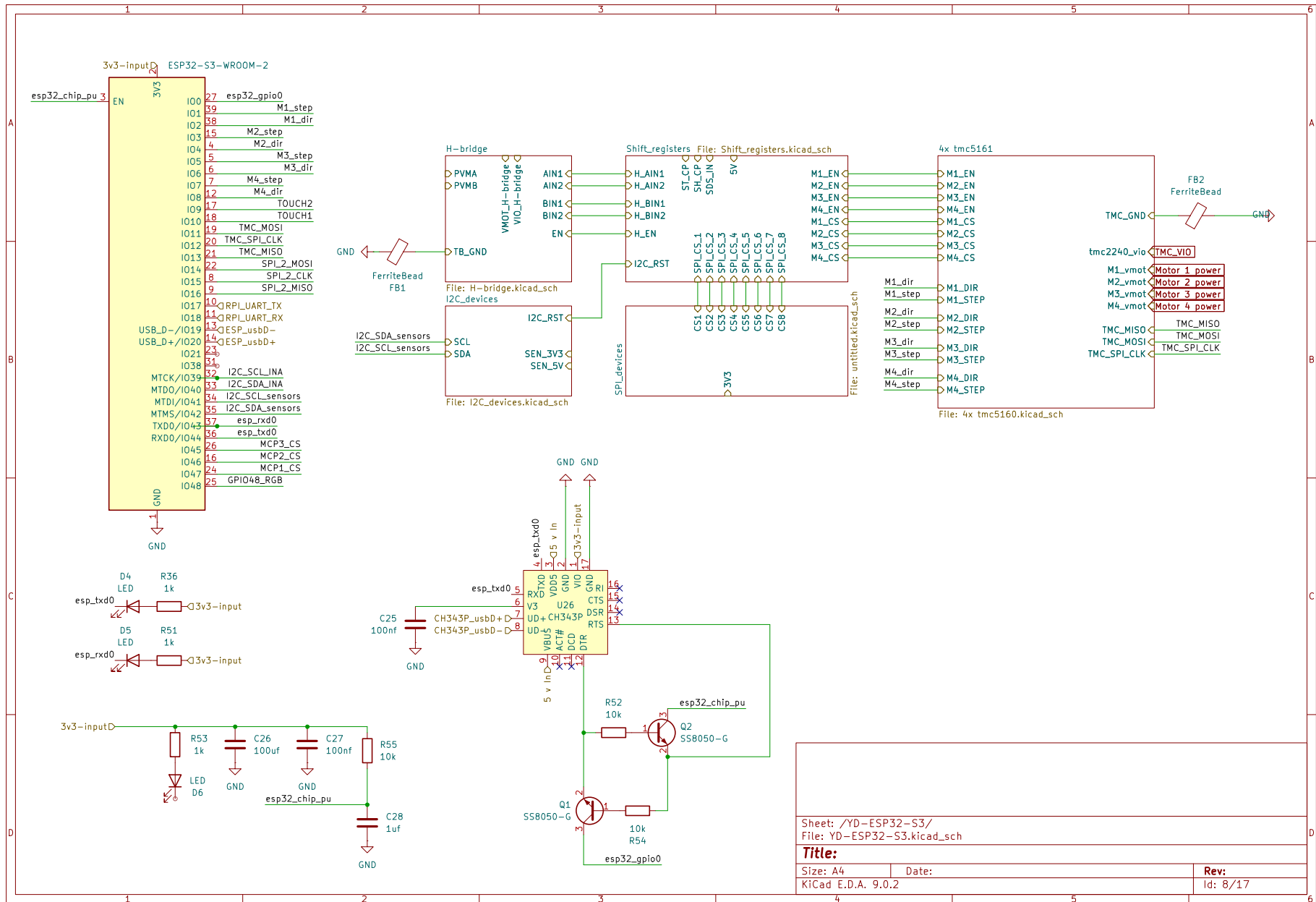
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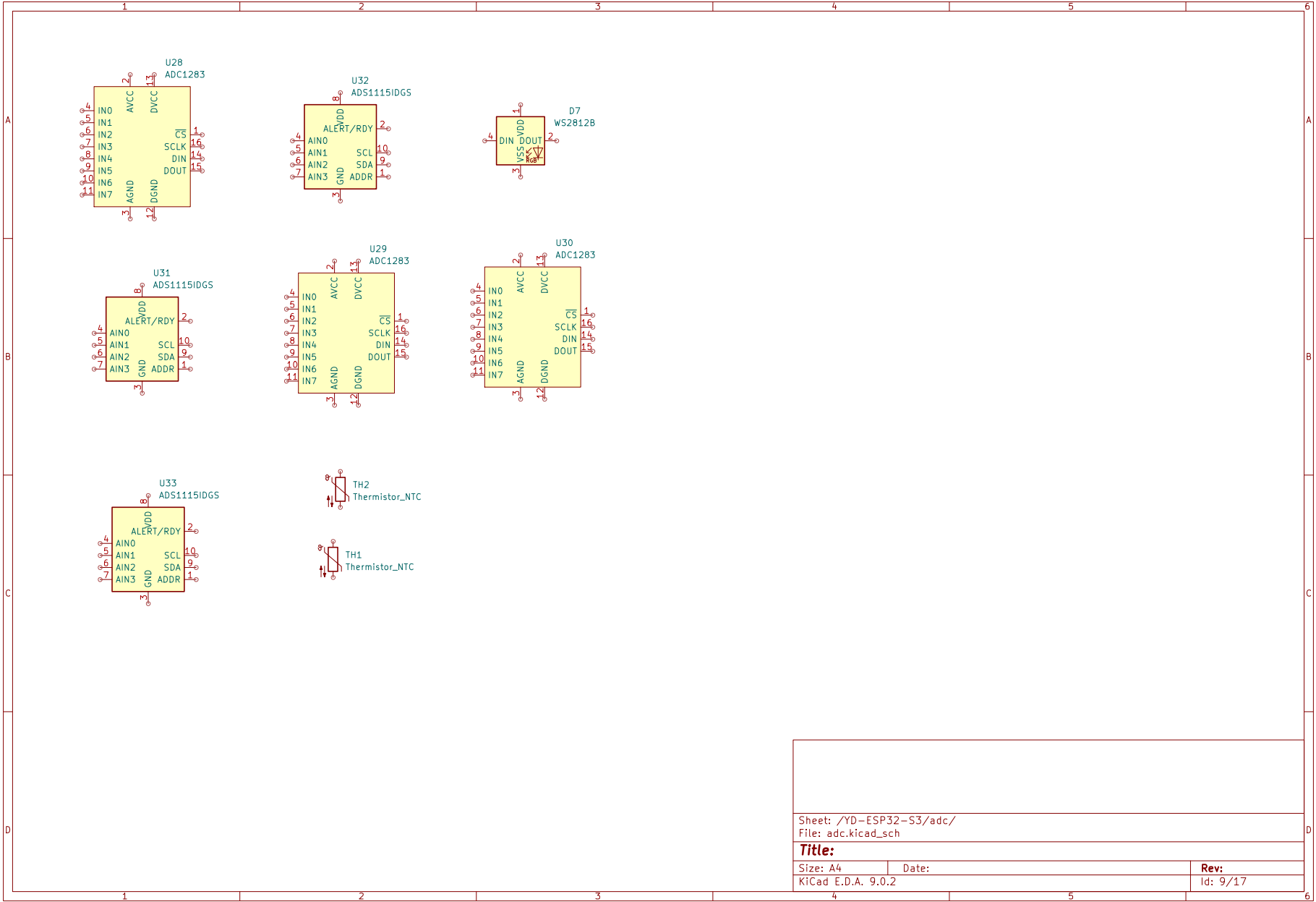
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Rev:

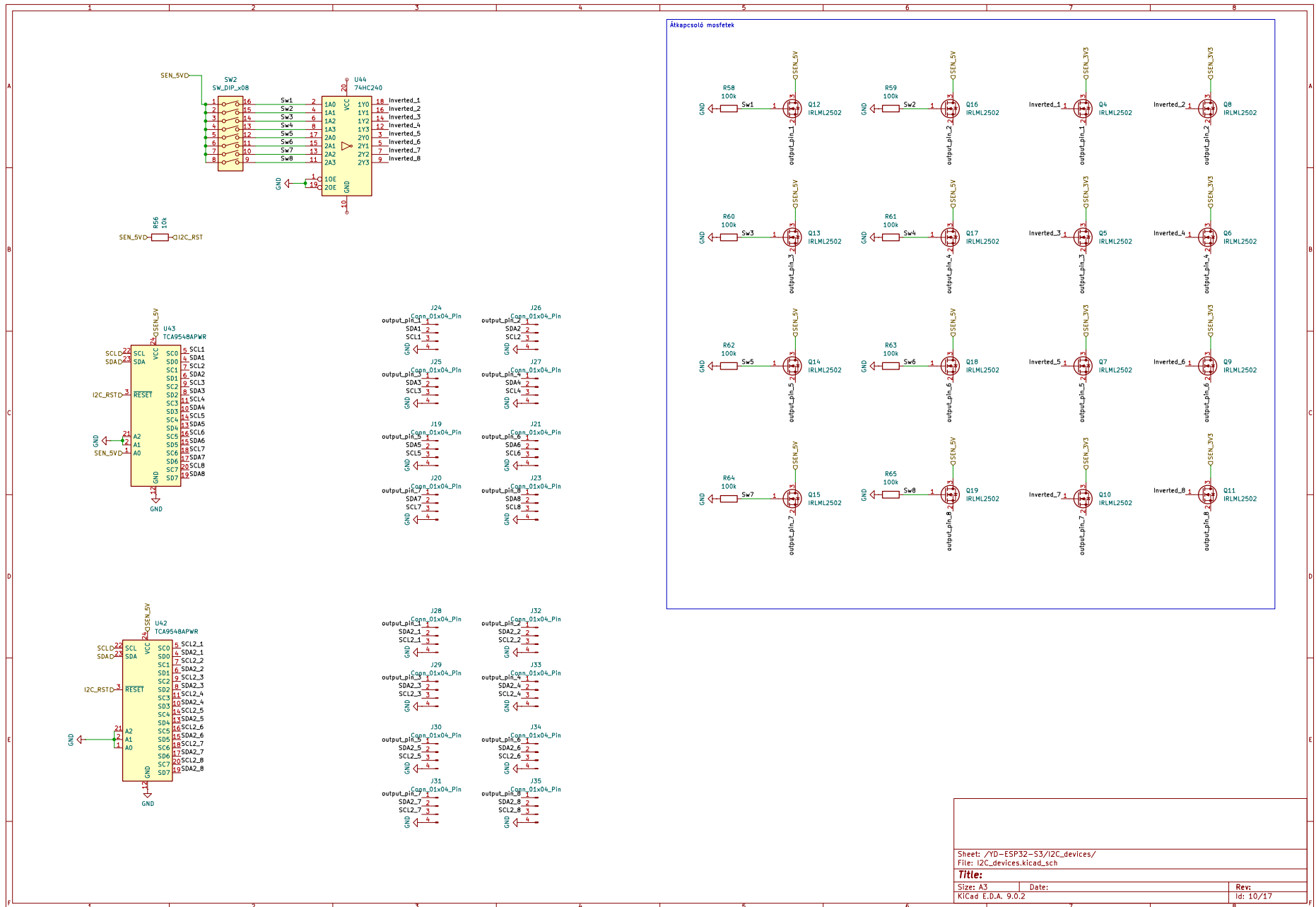
KiCad E.D.A. 9.0.2

Id: 7/17

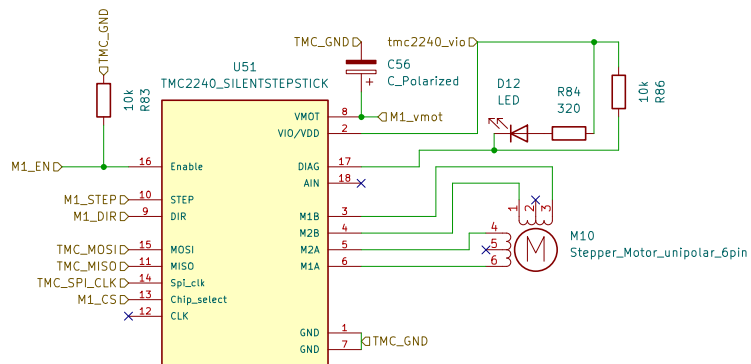




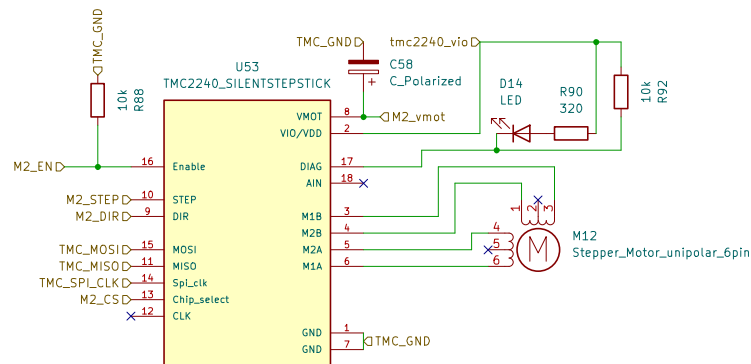
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Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.2		Id: 9/17



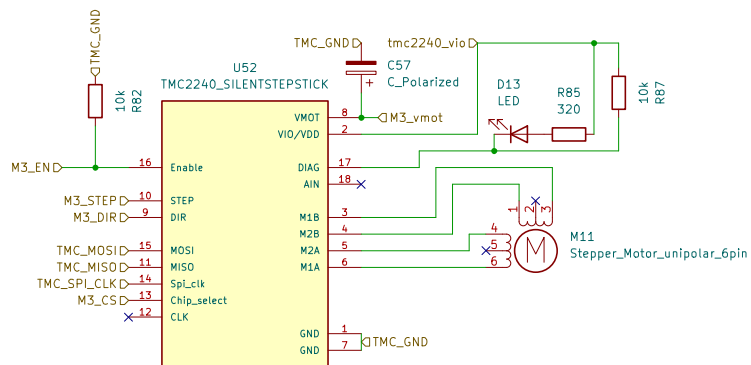
M1_control with TMC2240/TMC5160



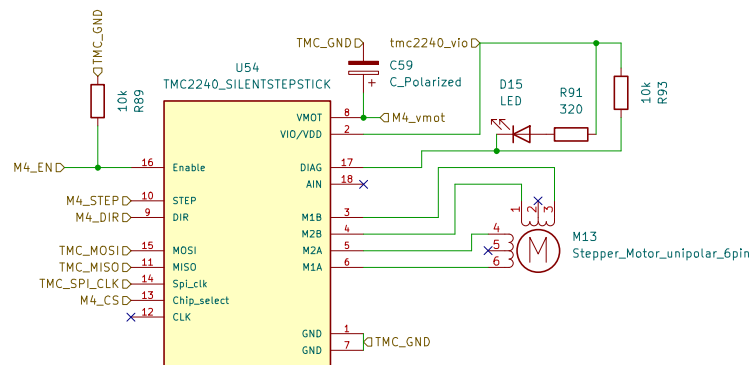
M2_control with TMC2240/TMC5160



M3_control with TMC2240/TMC5160



M4_control with TMC2240/TMC5160

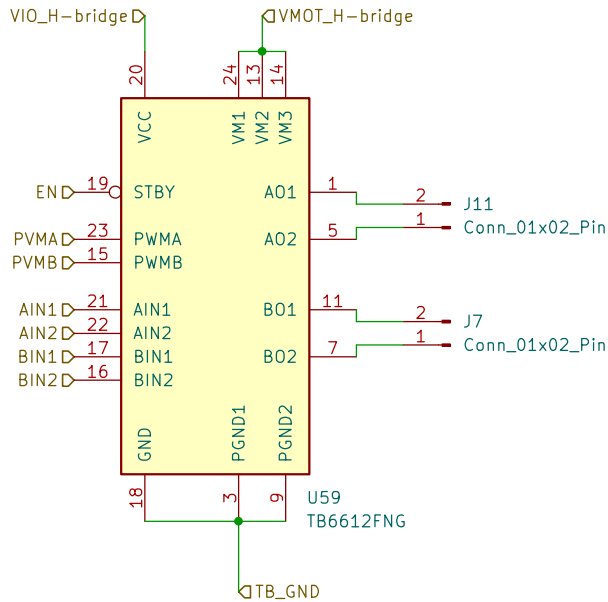


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File: 4x tmc5160.kicad_sch

Title:

Size: A4 Date:
KiCad E.D.A. 9.0.2

Rev:
Id: 11/17



```
//motor A connected between A01 and A02
//motor B connected between B01 and B02
```

```
int STBY = 10; //standby
```

```
//Motor A
int PWMA = 3; //Speed control
int AIN1 = 9; //Direction
int AIN2 = 8; //Direction
```

```
//Motor B
int PWMB = 5; //Speed control
int BIN1 = 11; //Direction
int BIN2 = 12; //Direction
```

```
void setup(){
  pinMode(STBY, OUTPUT);
```

```
  pinMode(PWMA, OUTPUT);
  pinMode(AIN1, OUTPUT);
  pinMode(AIN2, OUTPUT);
```

```
  pinMode(PWMB, OUTPUT);
  pinMode(BIN1, OUTPUT);
  pinMode(BIN2, OUTPUT);
}
```

```
void loop(){
  move(1, 255, 1); //motor 1, full speed, left
  move(2, 255, 1); //motor 2, full speed, left

  delay(1000); //go for 1 second
  stop(); //stop
  delay(250); //hold for 250ms until move again

  move(1, 128, 0); //motor 1, half speed, right
  move(2, 128, 0); //motor 2, half speed, right

  delay(1000);
  stop();
  delay(250);
}
```

```
void move(int motor, int speed, int direction){
  //Move specific motor at speed and direction
  //motor: 0 for B 1 for A
  //speed: 0 is off, and 255 is full speed
  //direction: 0 clockwise, 1 counter-clockwise
```

```
  digitalWrite(STBY, HIGH); //disable standby
```

```
  boolean inPin1 = LOW;
  boolean inPin2 = HIGH;
```

```
  if(direction == 1){
    inPin1 = HIGH;
    inPin2 = LOW;
  }
```

```
  if(motor == 1){
    digitalWrite(AIN1, inPin1);
    digitalWrite(AIN2, inPin2);
    analogWrite(PWMA, speed);
  }else{
    digitalWrite(BIN1, inPin1);
    digitalWrite(BIN2, inPin2);
    analogWrite(PWMB, speed);
  }
}
```

```
void stop(){
  //enable standby
  digitalWrite(STBY, LOW);
}
```

Sheet: /YD-ESP32-S3/H-bridge/
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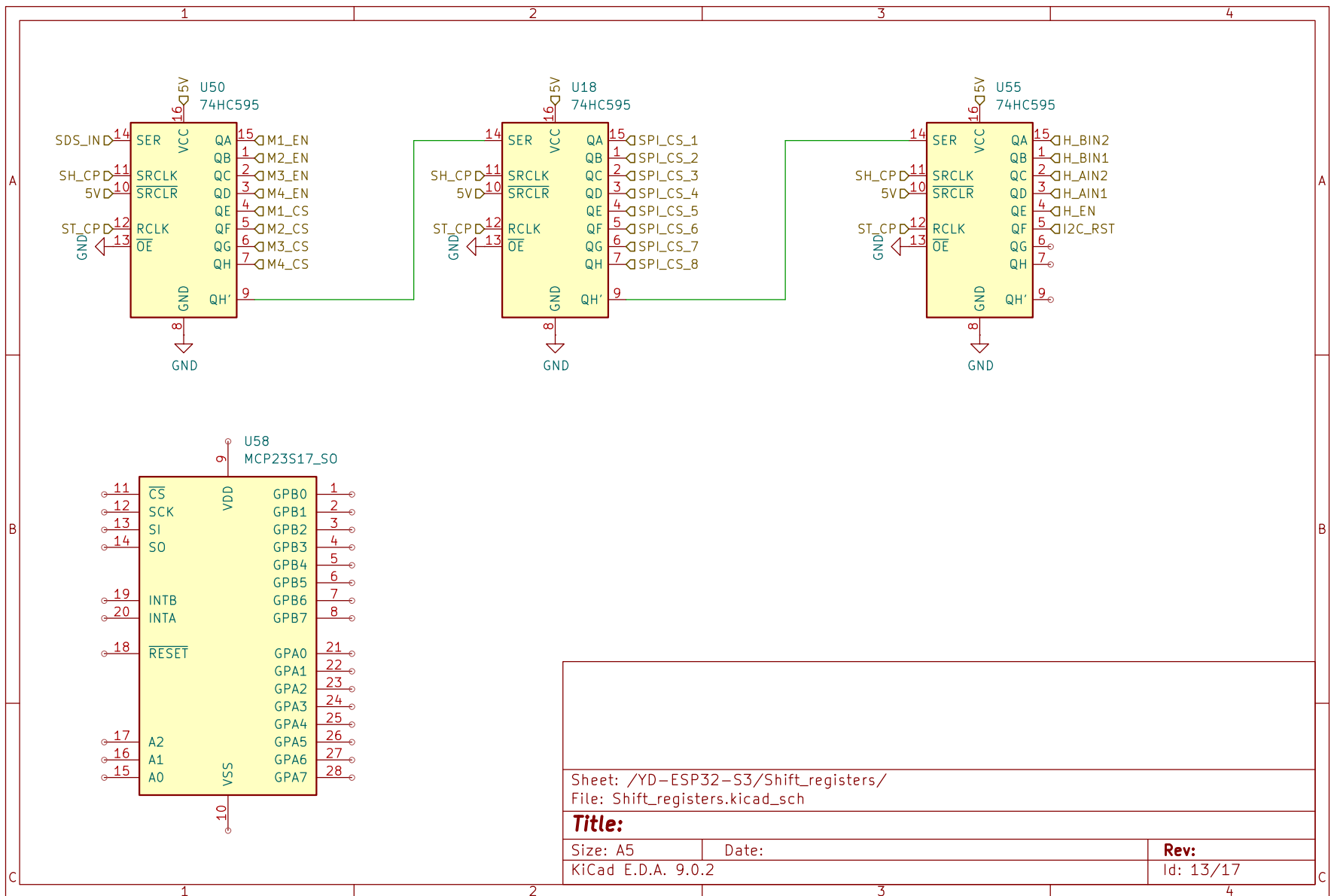
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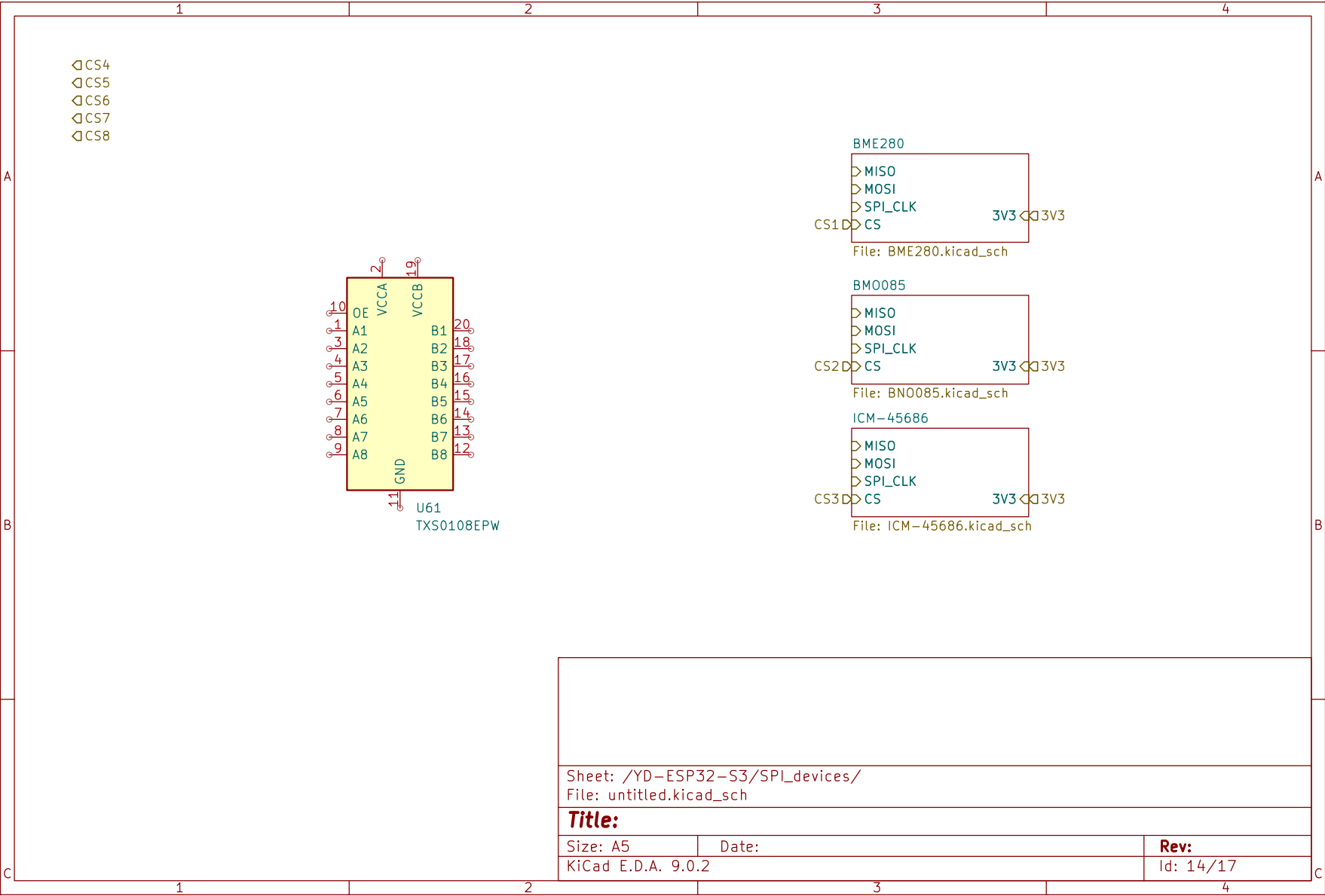
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Rev:

Id: 12/17





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Date:

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Id: 14/17

